



Flight Testing for Cosmic Radiation Detection on High-Altitude Balloons

Leo Cohen, Aaron Jia, Alissa Sperling



Background

Cosmic Rays and the Earth's Atmosphere

Energetic events across the universe emit cosmic rays that hurtle toward the Earth, colliding with the Earth's atmosphere. As these cosmic rays, composed primarily of protons, enter the Earth's atmosphere, they decay, creating showers of secondary cosmic rays that contain additional subatomic particles, including muons, pions, and various neutrinos. The Regener-Pfotzer maximum, around 20 km above Earth, marks the peak of secondary cosmic-ray flux before it decreases with altitude. These cosmic ray showers can be detected by simple particle detectors commonly used in ground-based experiments. While previously there have been some attempts to operate particle detectors at various altitudes, the research collaboration represents the first attempt to create a coordinated array of cosmic ray detectors flying from the surface of the Earth to over 30 km altitude across multiple latitudes.

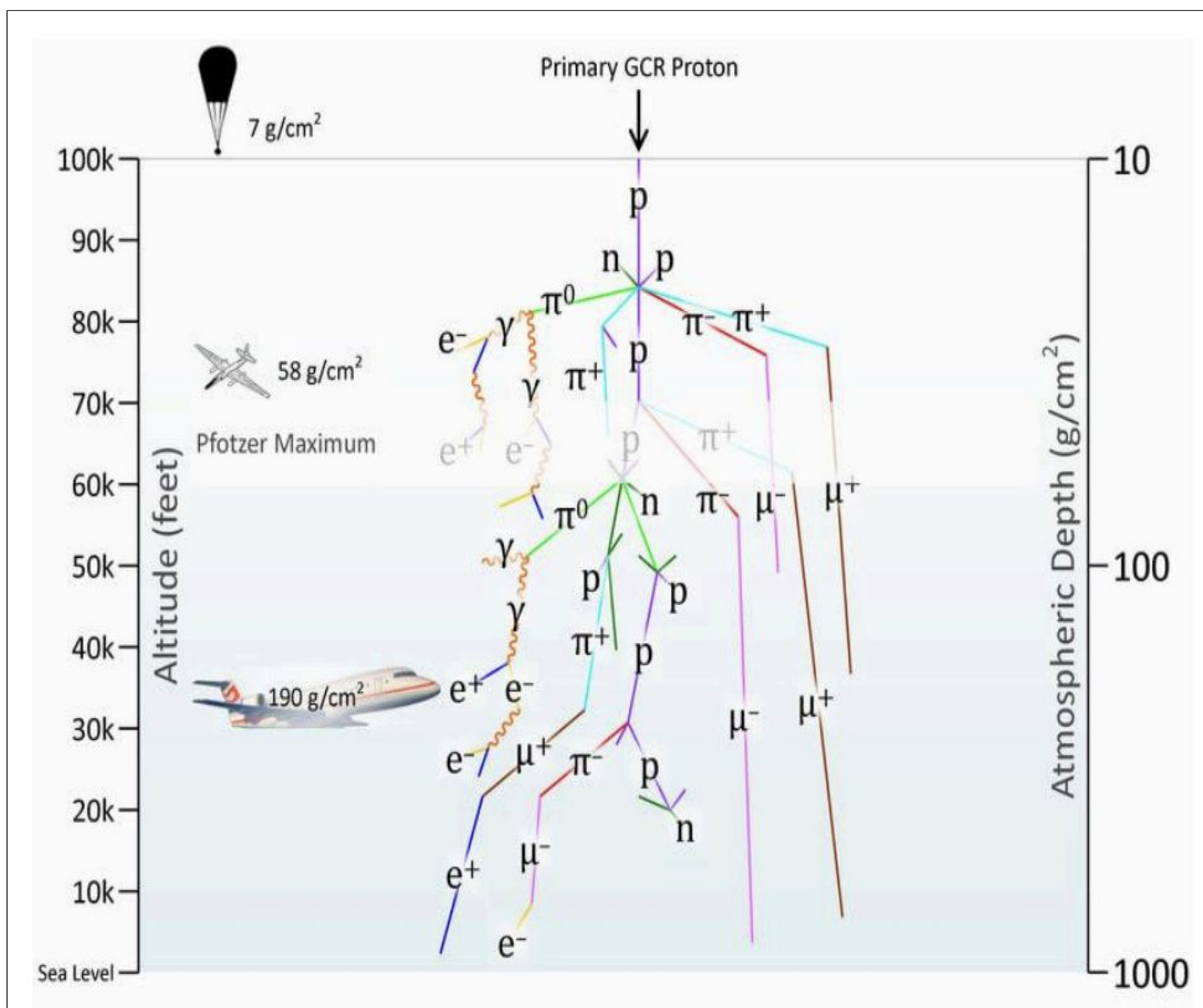


Figure 1. Primary cosmic rays trigger a cascade of secondary particles, including muons, which decay into an electron and neutrinos. The Regener-Pfotzer maximum occurs at 18-21 km altitude.



Figure 2. A high-altitude balloon bursting at an altitude of 26 km.

Purpose

- To evaluate the impact of muon detector scintillator plate orientation on particle detection rates.
- To test the function of muon detector detectors on high-altitude balloon flights with special consideration toward data storage, battery life, and detector function at low temperatures.
- To study the how subatomic particles interact with Earth's atmosphere and how the Regener-Pfotzer maximum varies based on latitude as a result of the Earth's magnetic field.

In-Flight Payload Stabilization

Particle Detector Tilt Testing

Scintillator-based particle detectors must remain parallel to the Earth's surface for accurate counts. To assess in-flight motion, tilt tests were conducted using Cosmic Watch and EduMOD particle detectors. Data aligned with the cosine squared law, showing decreased muon flux at greater angles. Horizontally traveling muons lose more energy and have a higher decay probability due to a longer atmospheric path. As expected, the EduMOD, with its larger scintillator, detected an average of 141 more particles per minute with under 5% variation at any angle.

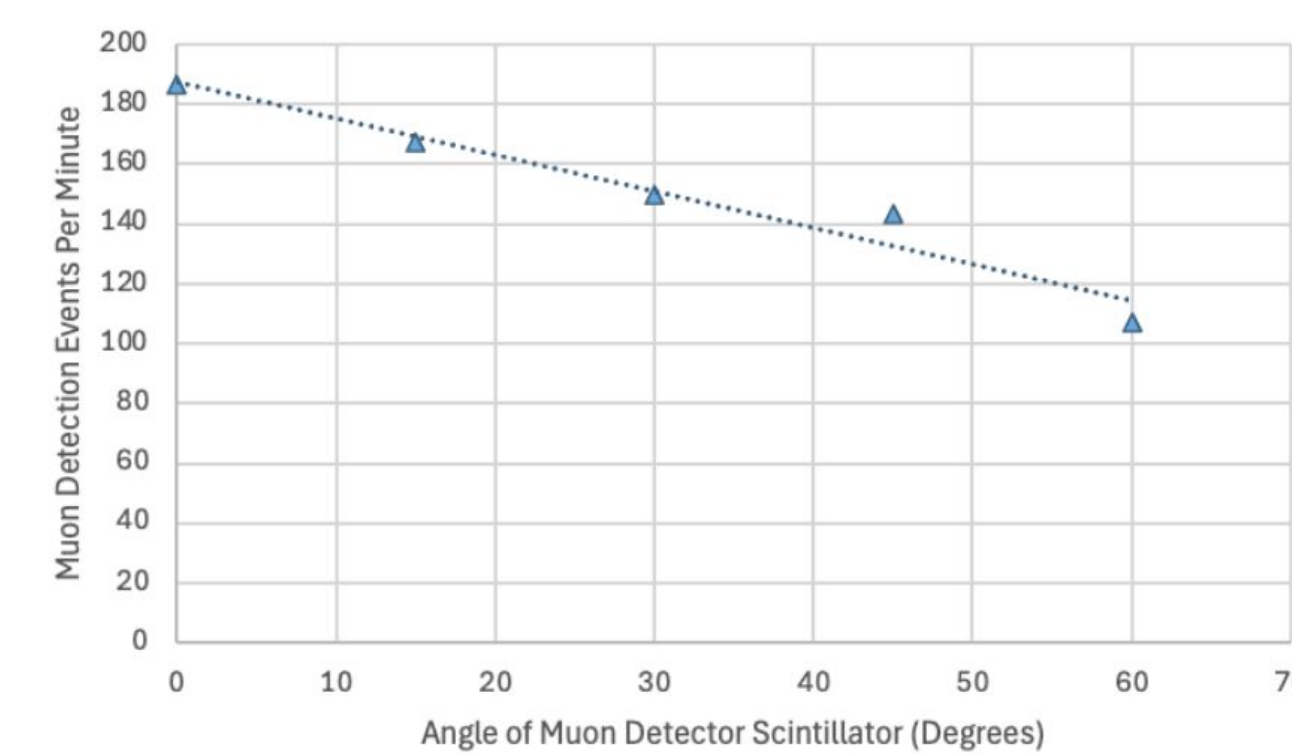


Figure 3. Average cosmic-ray muon detection events per minute as a function of the angle of the EduMOD muon detector scintillator. Detection events per minute decrease linearly at a rate of 1.21 per degree tilted with an R^2 value of 0.948.

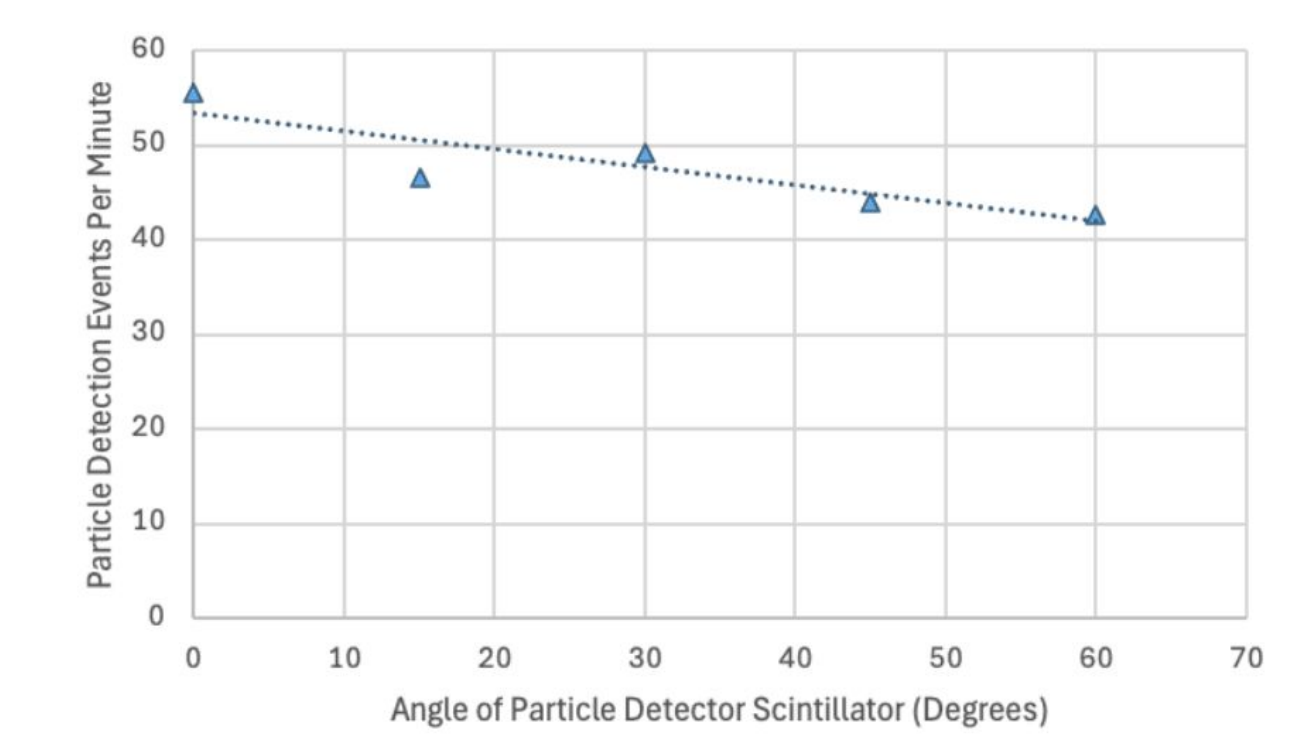


Figure 4. Average cosmic-ray particle detection events per minute as a function of the angle of the Cosmic Watch detector scintillator. Detection events per minute decrease linearly at a rate of .19 per degree tilted with an R^2 value of 0.776.

Payload Stabilization Experiments

A payload stabilization method was tested to reduce detector tilt in flight. The design used a modified passive plastic gimbal with a 3D-printed insert securing the Adafruit Data Logger, real-time clock, and accelerometer. A balloon launch compared a stabilized system to an unstabilized control. The stabilization system significantly reduced roll, from an average of 36.7° to 8.5°, and slightly improved pitch stability, lowering the average from 13.3° to 10.2°. This reduction in roll suggests the system effectively countered lateral motion, while its smaller impact on pitch indicates limitations in forward-backward stabilization.



Figure 5. Passive stabilization gimbal with the 3D printed insert and arduino accelerometer



Figure 6. Parallel payload system for the stabilized and unstabilized accelerometers.

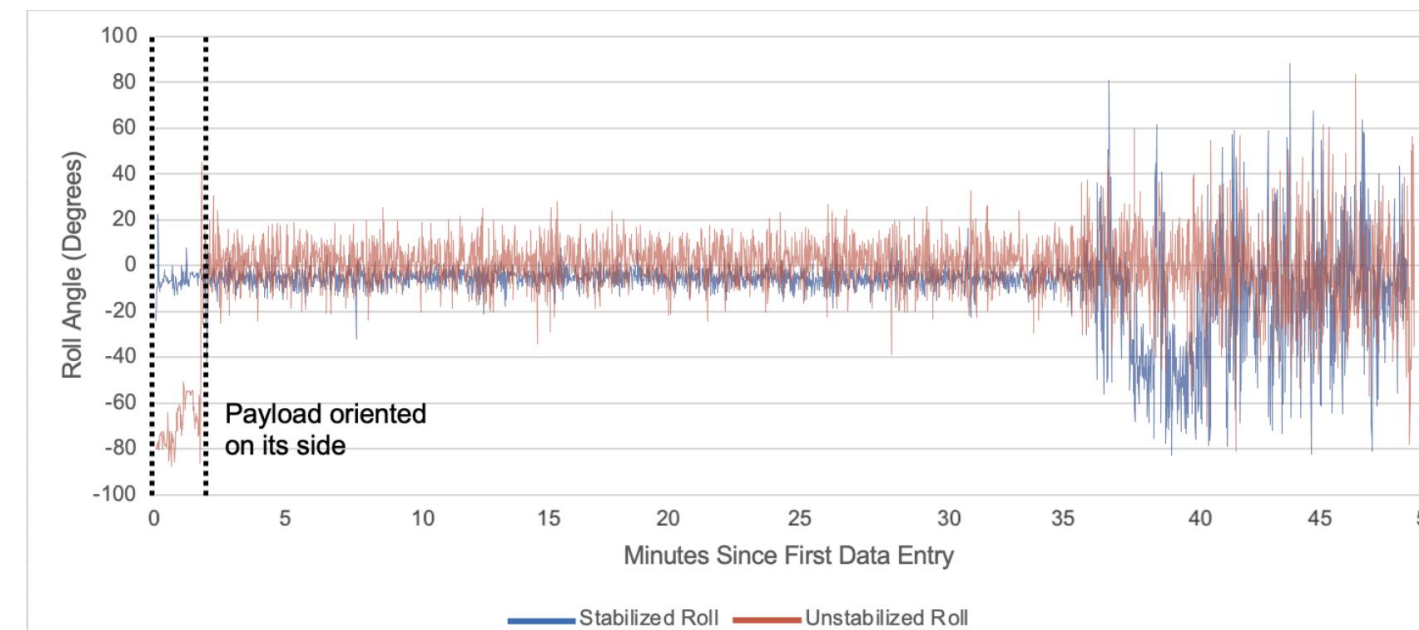


Figure 7. Stabilized vs unstabilized payload roll

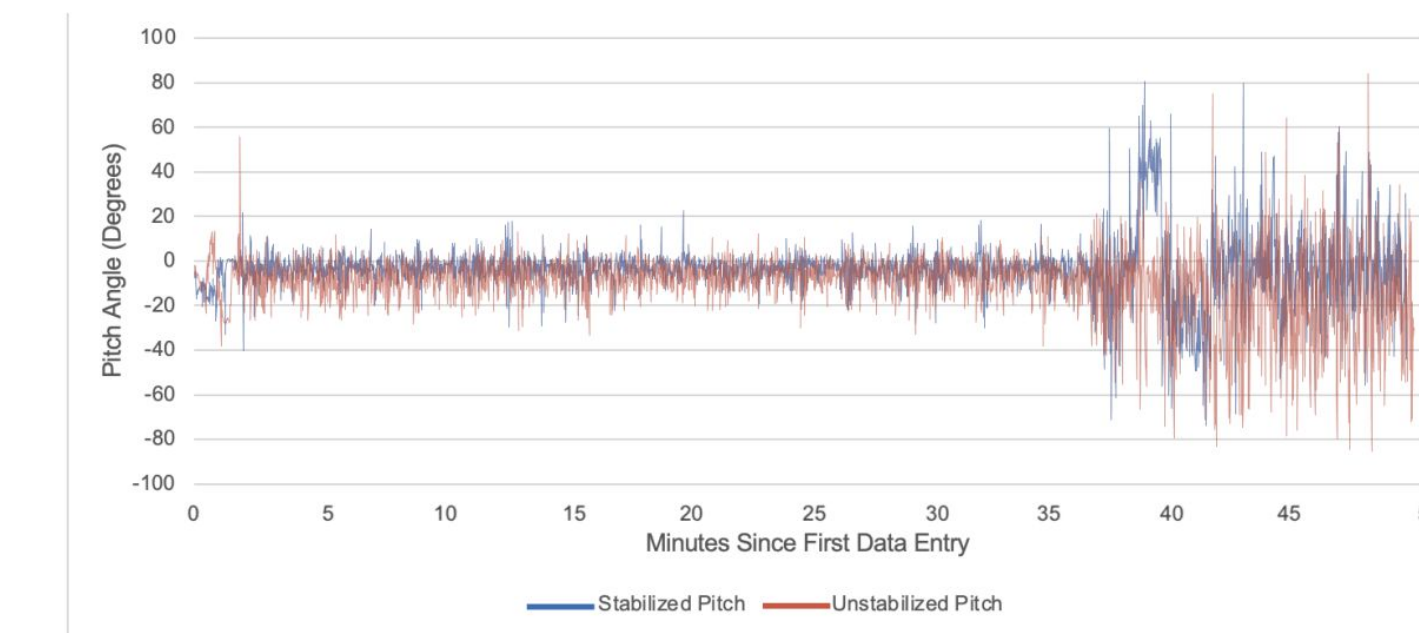


Figure 8. Stabilized vs unstabilized payload pitch

Preliminary Muon Detector Balloon Flight

Two Cosmic Watch particle detectors were launched on high-altitude balloons in October 2024, while an EduMOD muon detector collected ground data for baseline comparison and to control for cosmic ray fluctuations. The EduMOD functioned without issues, recording a stable muon rate averaging 200 events per minute (max: 241, min: 118, SD: 15.8). One Cosmic Watch detector experienced a memory leak, but the other successfully recorded data. Muon detections increased during ascent, peaking at 2,549 events per minute at ~19 km, aligning with the Regener-Pfotzer Maximum. Detection rates then declined, averaging 2,397 at 18-22 km, 2,167 at 22-24 km, and 1,940 at 24-26 km.

Preliminary Muon Detector Balloon Flight Events

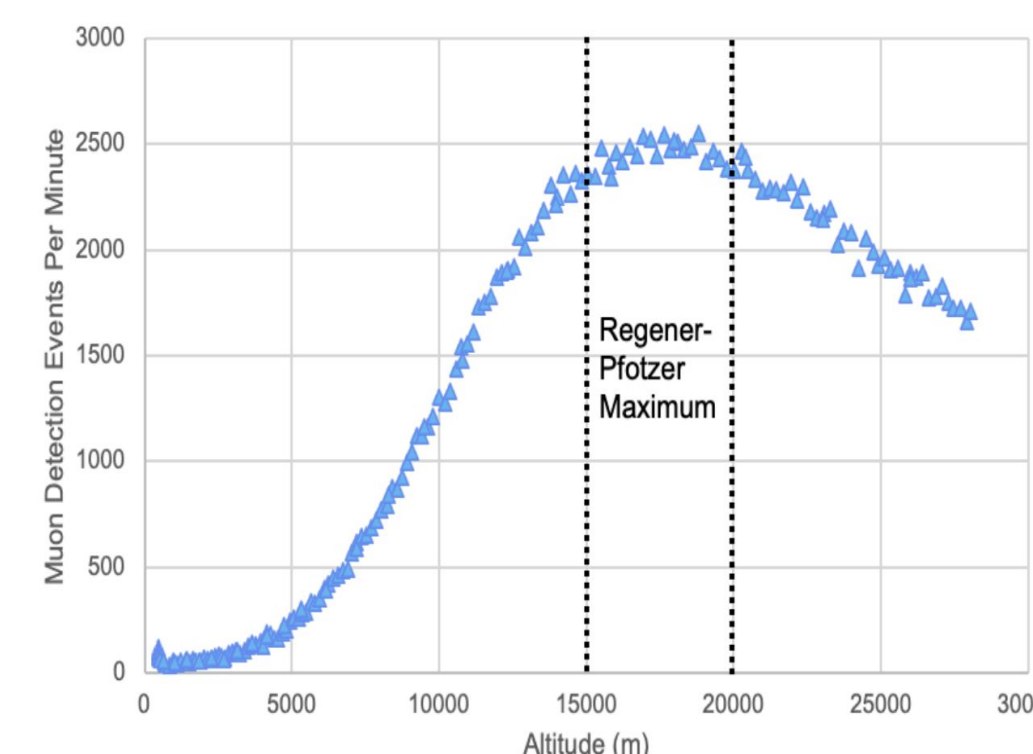


Figure 9. Muon detection events vs altitude from the Cosmic Watch during a high-altitude balloon flight.

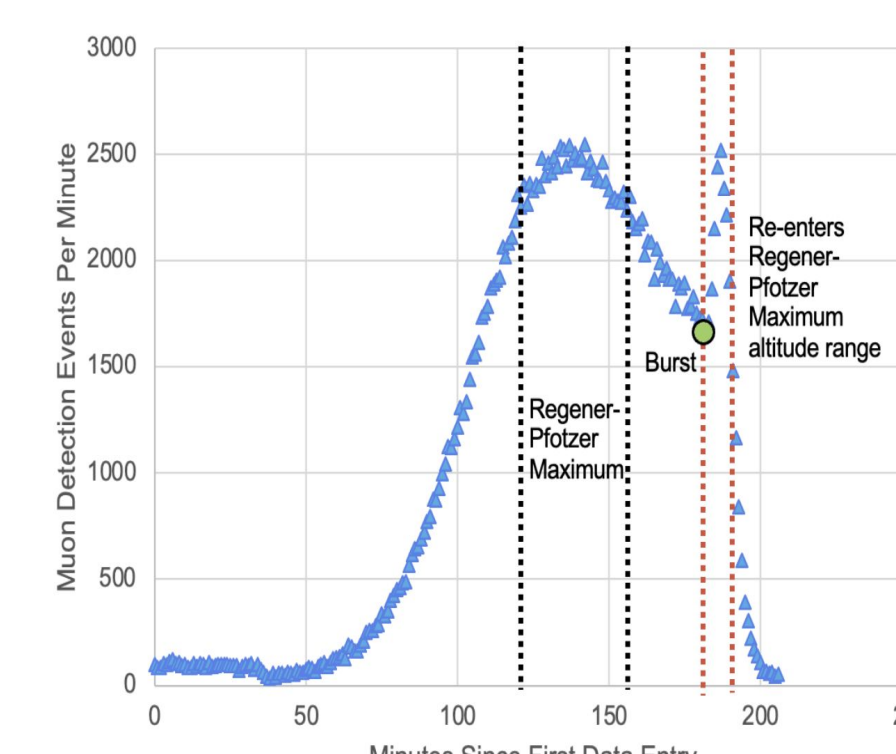


Figure 10. Muon detection vs time from the Cosmic Watch during a high-altitude balloon flight.

Stabilized Detector and EduMOD Launch

Stabilized Detector Testing

Based on the success of the fall 2024 stabilization tests, a custom stabilization gimbal was designed and fabricated for the Cosmic Watch in anticipation of the spring 2025 launches. The gimbal was printed from PLA and utilized hex bearings and 3/8-inch standoffs to enable smooth rotation on both the roll and pitch axes.

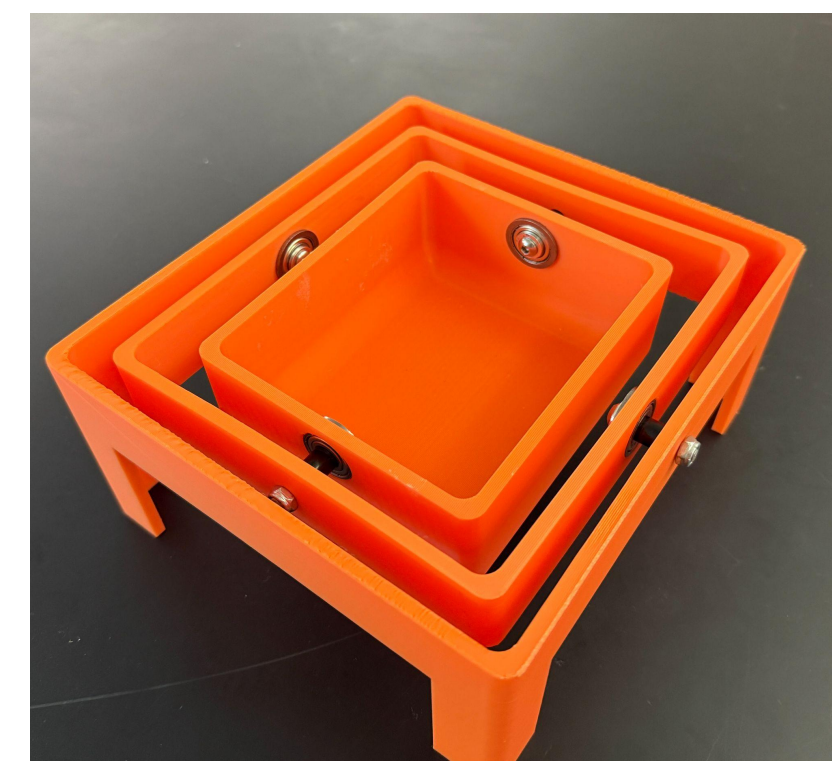
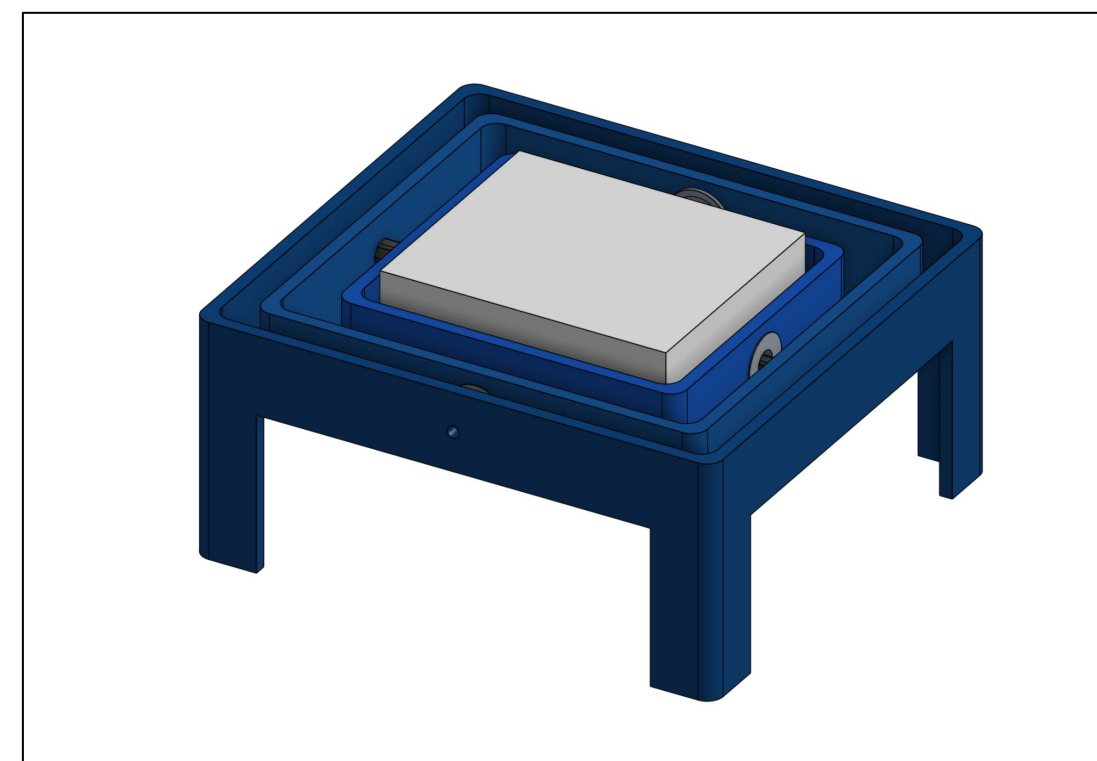


Fig. 11 & 12. The stabilized gimbal system rendered in a computer-aided design program and the 3D printed assembly.

The stabilized Cosmic Watch and an unstabilized Cosmic Watch were launched in tandem, and their respective particle detection rates during the duration of the launch were recorded. A Wilcoxon Rank-Sign test with continuity correction was conducted to determine the statistical significance of the paired differences in the recorded cosmic ray rates between the two detectors. A paired difference was calculated for every minute of the balloon launch using the formula [unstabilized Cosmic Watch cosmic ray rate - stabilized Cosmic Watch cosmic ray rate], with the highest rank being 121 and lowest being 1. The sum of ranks for the positive paired differences was compared to its expected value. The calculated p-value from the z-score was 5.285×10^{-10} , indicating the cosmic ray rates recorded by the stabilized Cosmic Watch were significantly greater than those recorded by the unstabilized Cosmic Watch.

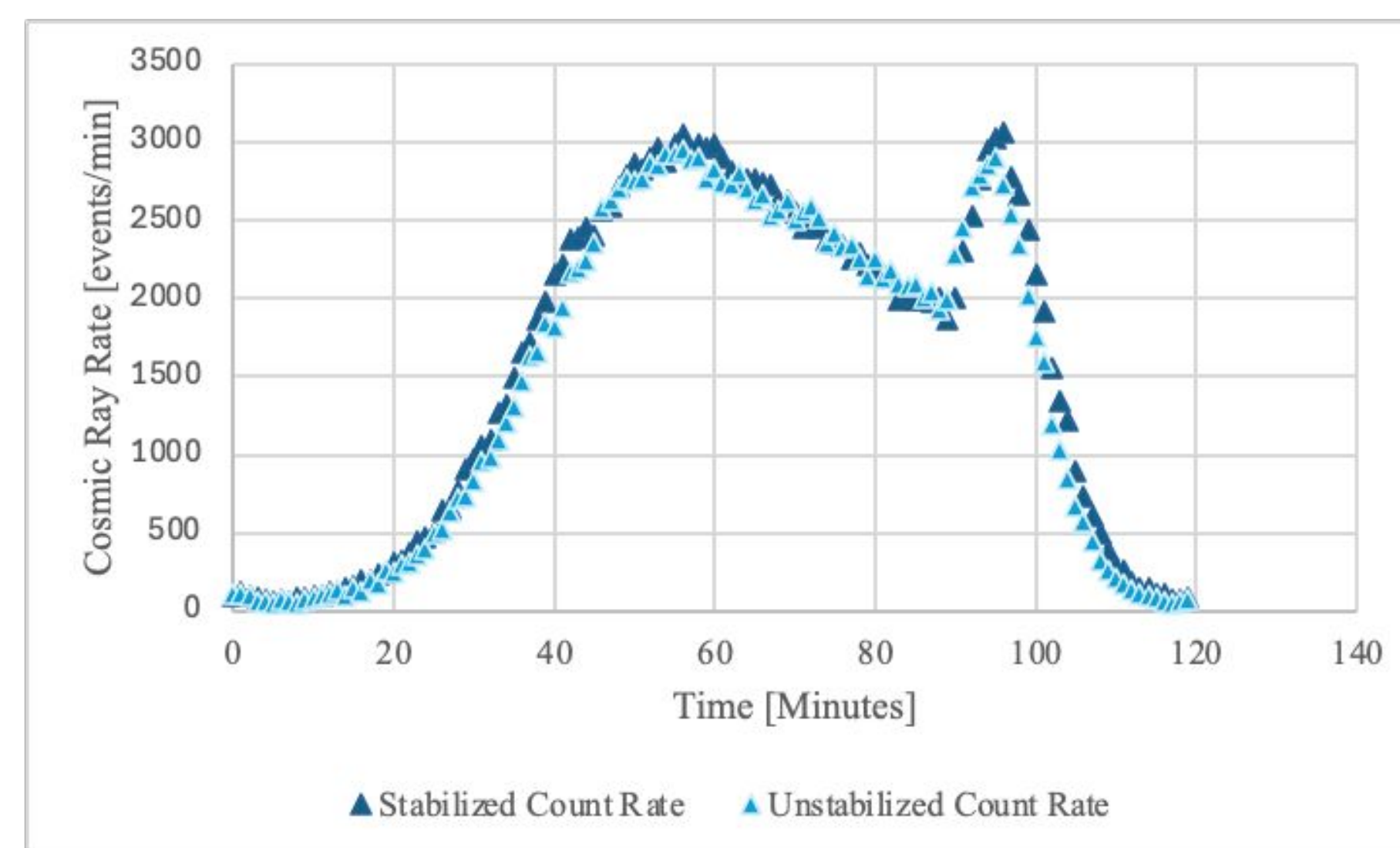


Fig. 13. Average cosmic ray rates measured by both the stabilized and unstabilized Cosmic Watches during the entire duration of the balloon launch. p-value = 5.285×10^{-10} .

Stabilized Detector and EduMOD Launch Continued

EduMOD Launch Results

An EduMOD muon detector was also launched in May of 2025. From approximately 85 minutes to 110 minutes into the flight, EduMOD's recorded count rate stayed relatively consistent, despite prior research and experimentation supporting an expected increase during this period. This indicates that the EduMOD likely became saturated during this time and consequently could not record a greater count rate than approximately 6,100 events per minute. However, the marked decrease in count rate after Maxima 1, prior to this period of potential saturation, is currently unexplained. As seen in Fig. 14, the detector reaches saturation at maxima 1 and then again after it. However, since the balloon is ascending until approximately minute 130, the EduMOD should still be saturated if that is what is occurring between minutes 85 and 110. The recorded muon count rate during the balloon flight indicates that further work is needed to ensure accurate measurements of muon flux with the EduMOD detector.

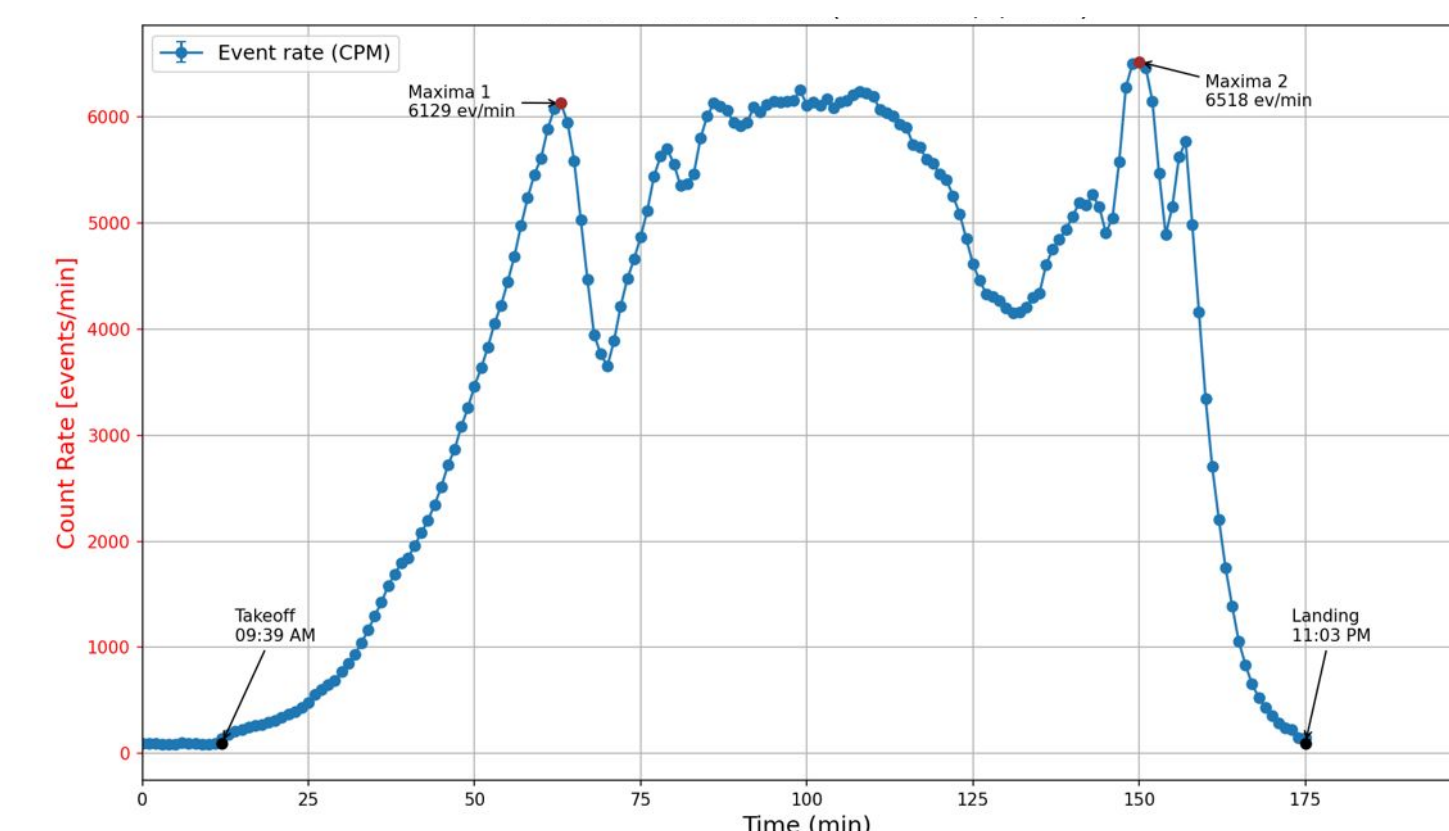


Fig. 14. Distribution of count rates recorded by the EduMOD muon detector during the duration of the balloon launch as a function of time.

Coordinated Particle Detector Array Pilot Launch

Concurrently with the spring 2025 detector launch, three other sites across the United States launched balloons equipped with particle detectors, all of which reached a peak altitude of 30 km at approximately local solar noon. A total of seven balloons were launched, with a combination of EduMODs, Cosmic Watches, and other sensors and telemetry devices distributed among the balloons. Institutions involved in the launches included Drexel University and Springside Chestnut Hill Academy in Pennsylvania, J.L. Mann High School and Furman University in South Carolina, Angelo State University and Wall High School in Texas, and Skyline College and The Bay School in California. All launches were conducted with students and faculty from both the secondary and higher education institutions. Mission planning, launches, and data analysis work was primarily conducted by students. This launch was the first attempt at a floating particle detector array, and future coordinated launches are planned for later in 2025.

Discussion and Future Work

Particle Detector Scintillator Orientation Matters

Muon counts decrease as the scintillator angle nears 90 degrees, aligning with prior research and highlighting the need for effective stabilization in high-altitude balloon flights.

Muon Detector Performance at Altitude

Preliminary tests showed peak detection rates near 19 km, consistent with the Regener-Pfotzer Maximum (18–21 km). These results confirm that Cosmic Watch functions reliably in near-space conditions.

Passive Stabilization Improves accuracy in Muon Count

A passive dual-axis gimbal was able to stabilize the Muon detector and shows the cosmic ray rates recorded by the stabilized Cosmic Watch were significantly greater than those recorded by the unstabilized Cosmic Watch.

Collaboration with HERA Project

In fall 2024, sixteen institutions formed HERA (High-altitude Engineering for Research in Astrophysics). HERA will conduct coordinated particle detector launches across the U.S. and Australia. The summer and fall 2025 launches will continue to study the Regener-Pfotzer maximum relative to Earth's magnetic field, with future experiments examining atmospheric conditions, the solar cycle impact on particle flux, as well as improving the data collected from the EduMOD detector.